CampusNotes - Academic Study Resource

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING - SEMESTER 5

OFFICIAL ACADEMIC STUDY MATERIAL - CSE CURRICULUM

Experiment 8: Floyd-Warshall All-Pairs Shortest Path Algorithm

Course: Design Analysis and Algorithms Laboratory [23CS3505]
Category: Experiments / Practicals | Experiment #8 | Academic Year: 2026-27

---

1. Overview & Objective:

Dynamic programming matrix relaxation algorithm computing all-pairs shortest distance paths in O V^3 time.

2. Practical Tasks & Verification Steps:

1. Initialize distance matrix D[V][V] with direct edge weights and infinity for non-adjacent pairs.
2. Execute 3 nested loops considering each vertex k as an intermediate candidate path vertex.
3. Apply relaxation D[i][j] = min D[i][j], D[i][k] + D[k][j] across all i, j pairs.
4. Detect negative weight cycles by inspecting diagonal entries D[i][i] < 0.

Submission Guidelines & Verification:

This document verifies end-to-end PDF storage, preview, and download for CampusNotes.
Administrators can safely edit metadata, publish/unpublish, or replace this document.